

Highlights

Process Mining and Stochastic Modeling for Software Process Forecasting: A Systematic Mapping Study with an LLM-Assisted Protocol

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- LLM-assisted mapping screens 6,147 papers across two corpus tiers.
- Human double-screening ($n = 468$ T/A, $n = 177$ FT) reports Lost Evidence.
- 340 studies after dedup; 1 jointly covers PM, stochastic, and forecasting.
- Five findings (F1–F5), SPMF taxonomy, and a four-direction agenda.
- Replication package on Zenodo (DOI 10.5281/zenodo.21939600).

Process Mining and Stochastic Modeling for Software Process Forecasting: A Systematic Mapping Study with an LLM-Assisted Protocol

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Abstract

Context. Software processes generate event data that process mining (PM) and stochastic models can use for forecasting, yet their integration for software development lifecycle (SDLC) prediction is reported only anecdotally. **Objective.** To map the state of the art at the intersection of process mining, stochastic modeling, and machine-learning forecasting applied to software processes; to identify research gaps; and to propose a research agenda. **Method.** We executed a systematic mapping study reported against SEGRESS [1], with LLM-assisted screening (*claude-haiku-4-5* as primary screener; *claude-sonnet-4-6* as verifier). Searches were executed on 12 April 2026 over January 1994–December 2025. From 5,783 deduplicated records plus 364 control items, two corpus tiers were screened: 2,340 working-set and 3,807 auxiliary papers (6,147 in total). Cohen’s κ was computed on stratified 20% LLM–LLM samples and on an independent blind human sample ($n = 468$ T/A, $n = 177$ FT). An 8-criterion reporting-quality checklist was applied to all confirmed includes. **Results.** 340 distinct primary studies were confirmed after cross-tier deduplication (404 raw confirmations: 169 working-set + 212 + 23 auxiliary). LLM–LLM agreement was substantial ($\kappa_{\text{binary}} = 0.695$ T/A, 0.694 FT); human–LLM agreement was fair at T/A (0.335) and slight at FT (0.122), with 75% Lost Evidence at full-text screening. Findings: F1 descriptive ceiling; F2 ad-hoc event-log construction; F3 weak stochastic–PM integration; F4 pipeline fragmentation (only 1 of 340 papers jointly satisfies PM, stochastic modeling, and forecasting); F5 process-derived ML features unexplored. **Conclusion.** The literature is descriptively rich but predictively fragmented. We propose the SPMF taxonomy and a four-direction research agenda. PATHCAST is sketched as a future-work hypothesis, not as an empirically validated pipeline. **Limitations.** Structural scarcity claims depend on an LLM selection protocol whose recall against a human gold standard remains limited at full-text screening; deduplication reduced the combined corpus from 404 to 340 studies.

Keywords: Systematic Mapping Study, Process Mining, Stochastic Modeling, Software Process Forecasting, LLM-Assisted Screening, Mining Software Repositories

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1. Introduction

This article reports a systematic mapping study conducted to map the state of the art at the intersection of process mining, stochastic modeling, and machine learning applied to software process forecasting. The review is reported against SEGRESS [1] and follows the guidelines established by Kitchenham and Charters [2] for evidence-based software engineering, complemented by the snowballing procedures described by Wohlin [3] and the mapping study guidelines of Petersen et al. [4]. To our knowledge, no prior systematic mapping or literature review has targeted the joint intersection of process mining, stochastic SDLC modeling, and ML-based process forecasting; the closest secondary studies address subsets of this space (process mining in agile development, software reliability modeling, or repository mining for prediction) without integrating all three technique families. The primary objective is to identify, classify, and count existing approaches that combine process-aware techniques with predictive modeling for software development outcomes, and to articulate the evidence gaps that motivate future integrated pipelines such as PATHCAST (Section 5.4).

The review addresses three overarching research questions, each decomposed into two sub-questions covering the application landscape of process mining in SDLC contexts, the techniques and prediction methods employed, and the degree of integration between process mining and stochastic forecasting methods.

2. Review Protocol

2.1. Research Questions

The SLR is guided by the following research questions, structured to progressively map the landscape from application scope to technique integration (mapping-study outputs: classification, frequencies, taxonomy):

2.2. Search Strategy

The search strategy combines automated database search with backward and forward snowballing. The automated search targets five digital libraries: IEEE Xplore, ACM Digital Library, Scopus, Web of Science, and SpringerLink.

2.2.1. PICO Decomposition

The search string construction follows the PICO framework adapted for software engineering, as shown in Table 2.

2.2.2. Search Strings

Three complementary search strings are executed to maximize coverage while maintaining precision.

Table 1: Research questions and expected outputs.

RQ	Sub-RQ	Question	Expected Output
RQ1	RQ1.1	Which SDLC phases have been analyzed with process mining?	Frequency distribution by phase
RQ1	RQ1.2	What event log sources are used and how are they constructed?	Taxonomy of sources and construction methods
RQ2	RQ2.1	Which process discovery algorithms have been applied to SDLC?	Comparative matrix
RQ2	RQ2.2	Which predictive and stochastic techniques have been used?	Technique classification
RQ3	RQ3.1	What is the level of integration between PM and stochastic methods?	Integration maturity assessment
RQ3	RQ3.2	What gaps exist for process-aware forecasting in the SDLC?	Gap analysis matrix

Table 2: PICO decomposition for search string construction.

Facet	Role	Concepts
Population	Application domain	Software engineering, SDLC, software process, DevOps, CI/CD
Intervention	Analytical technique	Process mining, Markov chain, Monte Carlo, predictive PM, ML
Comparison	N/A	(omitted)
Outcome	Result	Forecasting, prediction, lead time, cycle time, process model

Principal String ($P \wedge I$).. Block P captures the software engineering context:

```
("software engineering" OR "software development" OR "software process" OR "software lifecycle"  
OR "software life cycle" OR "SDLC" OR "development lifecycle" OR "development life cycle"  
OR "DevOps" OR "CI/CD" OR "continuous integration" OR "continuous delivery" OR "agile development"  
OR "agile process" OR "issue tracking" OR "bug tracking" OR "version control" OR "code review"  
OR "pull request" OR "software project" OR "software repository" OR "software maintenance"  
OR "software evolution")
```

Block I captures the process mining and stochastic modeling techniques:

```
("process mining" OR "process discovery" OR "conformance checking" OR "workflow mining" OR  
"predictive process monitoring" OR "process prediction" OR "remaining time prediction" OR  
"process forecasting" OR "event log" OR "process simulation" OR "Monte Carlo simulation" OR  
"Markov chain" OR "Markov model" OR "stochastic process model" OR "transition probability"  
OR "absorbing chain" OR "lead time prediction" OR "cycle time prediction")
```

Complementary String ($MSR \rightarrow Process\ Modeling$).. Captures studies using MSR terminology without canonical process mining terms:

```
("mining software repositories" OR "software repository mining" OR "GitHub mining" OR "Jira  
mining") AND ("process model" OR "Markov" OR "Monte Carlo" OR "stochastic" OR "lead time"  
OR "cycle time" OR "throughput" OR "transition matrix")
```

Frontier String ($Stochastic \wedge PM \wedge SE$).. Targets the intersection of stochastic methods with process mining in software contexts:

```
("Monte Carlo" OR "Markov chain" OR "stochastic simulation" OR "probabilistic forecast" OR  
"transition matrix") AND ("process mining" OR "event log" OR "workflow mining" OR "process  
model" OR "business process") AND ("software development" OR "software engineering" OR "software  
process" OR "DevOps" OR "SDLC" OR "issue tracker" OR "commit log" OR "build pipeline")
```

The strings are adapted to each database’s syntax and field conventions as summarized in Table 3. The eligibility window covers publications from January 1994 to December 2025 (full calendar years). All database searches were executed on 12 April 2026; publications dated 2026 that were already indexed at execution time were retained.

2.2.3. Snowballing Procedure

Backward snowballing examines the reference lists of the control-set seed papers (Section 2.2, Table 4). Forward snowballing uses Google Scholar citation tracking to identify subsequent works that cite those seed papers. Both rounds apply the same inclusion and exclusion criteria defined in Section 2.3.

Table 3: Database-specific search adaptations and raw retrieval counts.

Database	Search Field	Strings	Filters	Records
Scopus	TITLE-ABS-KEY	3	ar, cp; EN; 1994–2025; exec. 2026-04-12	2,377
IEEE Xplore	All Metadata	4 (split)	Conf. + Journals; 1994–2025	1,386
Web of Science	TS (Topic)	3	Article, Proc.; EN; 1994–2025	997
SpringerLink	Simplified	8 queries	Chapter, Article; EN; 1994–2025	2,884
ACM DL	Abstract	2	Date: 1994–2025	96
Snowballing	–	–	Backward + forward	595
Control set	–	–	Manual	12
Total				8,347

2.2.4. Search String Validation

The search strings are validated against a control set of 10 papers that are known to be relevant to the review scope (Table 4). The acceptance threshold is $\geq 8/10$ papers found; the validation returned 10/10 (100%, PASS).

2.3. Inclusion and Exclusion Criteria

The inclusion criteria are organized in two tiers. Formal criteria (IC1–IC3) filter on publication attributes and are applied during the database search configuration. Content criteria (IC4a–IC4d) require that the study’s research contribution addresses at least one of four specific technique-domain intersections; they are applied during title/abstract and full-text screening. Exclusion criteria (EC1–EC5) are applied cumulatively across both screening phases. Criteria EC1–EC4 are evaluated by the LLM-assisted screener; EC5 (full text inaccessible) is applied programmatically after the inter-library-loan recovery pass for Band D/E papers fails to retrieve a copy. In the auxiliary tier, EC5 is extended to cover non-recoverable metadata insufficiency: papers for which no usable abstract could be recovered after the multi-source enrichment cascade are flagged for human review in the revised protocol. In the **original execution run**, papers for which the primary and verifier raters produced an irreconcilable decision without accessible full text were closed under EC5 rather than escalated to human arbitration; Section 6 quantifies the resulting Lost Evidence against an independent human gold standard, and the revised protocol treats T/A maybe and FT pending as positive referrals (LLM4SCREENLIT [5] R6).

At least one of IC4a–IC4d must be satisfied for inclusion. Criterion IC4b ensures that seminal works applying Markov chains to software processes—such as Whittaker and Thomason [6]—are captured despite predating the term “process mining.” To avoid colliding with the publication filters IC1–IC3, Results and Appendix A report the four content families as **TF-PM**, **TF-ST**, **TF-FC**, and **TF-MSR** (IC4a–IC4d, respectively).

Table 4: Control papers for search string validation.

#	Paper	Justification	Captured via
V1	Cook & Wolf (1998)	Seminal PM + SE	ACM
V2	Rubin et al. (2007)	PM framework for SW processes	Scopus
V3	Poncin et al. (2011)	PM + MSR intersection	IEEE
V4	Caldeira & Cardoso (2019)	PM + SW process assessment	IEEE
V5	Lemos et al. (2011)	PM for SW process management	IEEE
V6	Whittaker & Thomason (1994)	Markov + SW testing (seminal)	Scopus
V7	Tax et al. (2017)	PPM seminal (BPM baseline)	Control
V8	Rogge-Solti & Weske (2015)	Stochastic + PM forecasting	Scopus
V9	Rozinat et al. (2009)	PM → Simulation bridge	Control
V10	Gupta & Sureka (2017)	PM multi-repository in SE	ACM

2.4. LLM-Assisted Screening Methodology

Given the scale of the working set (2,340 papers) and the requirement for systematic, reproducible screening decisions with traceable rationale, the title/abstract screening phase employs a language model as the primary screener. Specifically, *claude-haiku-4-5-20251001* is used via the Anthropic Message Batches API, with a structured prompt encoding the IC4a–IC4d and EC1–EC4 criteria (EC5 is applied programmatically post-screening), the PATHCAST scope description, and decision rules for the three-class output: *include*, *exclude*, or *maybe*.

The screening prompt instructs the model to (i) identify which ICs are satisfied, (ii) identify which ECs apply, (iii) assign a decision with a one-to-two sentence rationale, and (iv) return the result as structured JSON. The *maybe* class captures genuine uncertainty—typically arising from missing abstracts or ambiguous domain terminology—without forcing a binary decision.

At the full-text screening phase, the same model is applied with a stricter prompt that enforces a binary outcome (*include* or *exclude*) with *pending* reserved only when the abstract is genuinely insufficient for a decision.⁴

This approach extends prior work on LLM-assisted systematic reviews [7, 8, 5] to the software process mining domain. The primary screener produces structured outputs that enable post-hoc quality analysis. Validation follows a three-stage protocol: (i) **independent human double-screening** on a stratified random 20% sample blind to the

⁴The two protocols use different nomenclature for the uncertainty class: T/A uses *maybe* (epistemically broad: criterion ambiguity *or* abstract insufficiency), whereas FT uses *pending* (narrowly: abstract insufficiency only). The semantic difference is intentional — the T/A pass operates on metadata where ambiguity is expected, while the FT pass commits to binary decisions whenever the abstract permits and escalates only the abstract-insufficient subset to manual full-text retrieval. This mapping is preserved in the cross-model κ analysis (Tables 16 and 17).

Table 5: Inclusion and exclusion criteria.

Type	ID	Criterion
Inclusion	IC1	Publication period: 1994–2025 (from Whittaker & Thomason onward).
	IC2	Language: English.
	IC3	Peer-reviewed journal article, conference paper, or workshop paper (≥ 4 pages).
	IC4a	Applies process mining techniques (discovery, conformance checking, predictive process monitoring) to software development artifacts (commits, issues, pull requests, CI/CD logs, bug trackers, VCS).
	IC4b	Uses stochastic models (Markov chains, Monte Carlo, stochastic Petri nets, absorbing chains) to analyze software development processes or workflows.
	IC4c	Proposes or evaluates forecasting of software process metrics (lead time, cycle time, remaining time, throughput, defect rates) from event logs or repository data.
	IC4d	Mines software repositories (GitHub, Jira, VCS, CI/CD platforms) to discover, analyze, or improve software process models.
Exclusion	EC1	Application domain is exclusively outside software development (healthcare, manufacturing, supply chain) without demonstrated relevance to SDLC processes.
	EC2	Software appears only as the implementation platform; the process being analyzed is not a software development process.
	EC3	Purely algorithmic or theoretical contribution without empirical evaluation in a software development context.
	EC4	Secondary study (SLR, mapping study, survey) not specifically addressing PM or stochastic methods in software engineering.
	EC5	Full text or decision-critical metadata inaccessible despite all retrieval attempts (open access, institutional, ILL, author-provided preprint, multi-source abstract enrichment). Applied programmatically after Band D/E recovery and auxiliary enrichment; also closes irreconcilable ambiguous cases when no accessible full text exists.

LLM’s decisions ($n = 468$ T/A, $n = 177$ FT), with Recall, Lost Evidence, and confusion-matrix reporting against the human rater as gold standard (Tables 14 and 15); (ii) **cross-model verification** on a separate stratified 20% sample, with *claude-sonnet-4-6* as second rater and Cohen’s κ reported following Landis and Koch (1977) [9]; and (iii) **author inspection** of all LLM include decisions before corpus freeze. No independent second human reviewer participated outside the blind sample (SEGRESS item 23c limitation). Cross-model verification is reported as a secondary, lower-cost signal; human-vs-LLM agreement is the primary validity evidence for selection outcomes.

Operationally, the **original execution run** did not preserve an open-ended manual-arbitration queue: unresolved maybe/pending cases were closed under EC5-extended. The **revised protocol** routes such cases to human review instead. Human validation (Section 6) shows that, under the original closure rule, all 54 T/A false negatives in the human sample were maybe rather than exclude decisions—a protocol artefact with measured Lost Evidence rather than hard LLM exclusions at T/A.

2.5. Reporting-Quality Assessment

Under SEGRESS [1] §4.3.2, risk-of-bias assessment is optional for mapping studies; we retained a **reporting-quality** checklist (not a risk-of-bias instrument) adapted from Dybå and Dingsøyr [10], with eight binary criteria. Its purpose is a sensitivity filter on synthesis tables (studies below the cut are excluded from the F1–F5 evidence base but remain in the confirmed set for context), not a validity judgement on primary-study outcomes:

Table 6: Quality assessment checklist.

#	Criterion	Score
QA1	Are the research objectives clearly stated?	0/1
QA2	Is the software engineering context described in detail?	0/1
QA3	Is the data source (event log) described reproducibly?	0/1
QA4	Is the PM or stochastic technique formally defined?	0/1
QA5	Are results validated empirically?	0/1
QA6	Are threats to validity discussed?	0/1
QA7	Are data and/or code publicly released?	0/1
QA8	Are process model quality metrics reported (fitness, precision)?	0/1

Studies scoring below 4.0/8 (50%) are excluded from the F1–F5 synthesis base. QA8 reflects whether studies report standard process-model quality metrics, not a framework-specific requirement.

QA scoring is performed by the same LLM-assisted protocol used for screening (Section 2.4): *claude-haiku-4-5-20251001* receives a strict rubric for each criterion and returns a binary score with a brief justification. The full prompt and the per-study scores are part of the replication package (Section 6.5). Table 7 reports the quality-assessment outcome for the complete confirmed set ($n = 169$). All 169 studies were scored.

Table 7: Quality-assessment results for the confirmed included set ($n = 169$; full QA applied).

QA Outcome	Count	Notes
Studies assessed	169	Absolute count and 100.0% of included set
Studies with score $\geq 4/8$	136	Retained for synthesis
Studies with score $< 4/8$	33	Excluded after QA
Mean QA score	4.88	Mean and standard deviation (1.40)
Median QA score	5.00	Median and interquartile range (4.00–6.00)

The 33 studies below the QA threshold are predominantly short workshop papers, position papers, or extended abstracts that fail QA3 (reproducible data source) and QA4 (formally defined technique) simultaneously. They are not removed from the PRISMA flow; they are retained in the included set as context but are not used in the F1–F5 evidence base reported in Section 5.1.

2.6. Data Extraction

For each included study, the attributes in Table 8 are extracted.

Table 8: Data extraction form.

Field	Description
ID	Unique identifier
Authors, Year, Venue	Bibliographic metadata
SDLC Phase(s)	Lifecycle phases covered
Event Log Source	Jira, GitHub, GitLab, Jenkins, etc.
Event Log Construction	Manual, semi-automatic, or automatic
PM Technique Category	Discovery / Conformance / Enhancement / Prediction
Specific Algorithm(s)	Alpha, Heuristics, Inductive Miner, etc.
Stochastic Method	Markov / Monte Carlo / Stochastic Petri Net / None
ML Technique	RF, XGBoost, LSTM, etc. / None
Prediction Target	Remaining time, outcome, lead time, delay, etc.
Integration Level	L0 None / L1 Sequential / L2 Pipeline / L3 Unified
Validation Type	Case study / Experiment / Industrial / Simulation
Tool/Platform	PM4Py, ProM, Disco, custom, etc.
Dataset Size	Number of events and cases analyzed
Process Model Quality	Fitness, precision, generalization (if reported)

3. Search Results and Study Selection

3.1. Database Retrieval and Deduplication

The three search strings executed across five databases returned 7,740 records. Backward and forward snowballing from the control set yielded an additional 595 records. Together with 12 control papers maintained as a validation-only set, the initial corpus totaled 8,347 records. Table 9 summarizes the retrieval by source.

Table 9: Record retrieval by source.

Source	Raw Records	After Internal Dedup
Scopus (6 queries)	2,377	1,852
SpringerLink (8 queries, 10 export batches)	2,884	1,543
IEEE Xplore (manual)	1,386	1,263
Snowballing	595	575
Web of Science (4 queries)	997	514
ACM Digital Library (3 queries)	96	34
Control set (validation only)	12	2
Total	8,347	5,783

Cross-source deduplication removed 2,564 duplicate records (30.7%), yielding 5,781 unique primary-corpus papers plus 2 unique control papers. Deduplication used normalized DOI matching, followed by fuzzy title matching (Levenshtein similarity ≥ 0.9) for records without DOIs.

3.2. Operational Working Set Construction

Screening 5,783 records in a single pass would require substantial resources. A working set of 2,340 papers was therefore constructed from the highest-signal primary queries—those whose query design directly targets the PATHCAST intersection (process mining, stochastic modeling, and software development)—as shown in Table 10. The working-set construction also incorporates 364 additional records from the manually imported control set and supplementary IEEE/ACM batches that were not part of the primary deduplication pool. The 3,807 unique records not selected for the working set therefore constitute the *high-recall auxiliary corpus*, retained as a frozen baseline and available for supplementary analysis if gaps in the primary results warrant expansion.⁵

The working set preserves 10/10 control papers when the validation-only control set is included, and 8/10 from primary queries alone—both above the 80% threshold.

⁵Composition: 5,783 unique deduplicated records, of which 1,976 were selected into the working set, plus 364 control and manually imported records added directly to the working set, totalling 2,340 working-set papers; the remaining 3,807 unique records constitute the auxiliary corpus.

Table 10: Working set composition.

Database	Queries included	Unique papers
Scopus	scopus_principal, scopus_complementar, scopus_frenteira, scopus_markov_testing, scopus_stochastic_petri, sco- pus_simulation_bridge	2,197
IEEE Xplore	ieee_manual_01 (PM + SE subgroup)	97
ACM DL	acm_principal, acm_msr	46
Total		2,340

3.3. Title/Abstract Screening

All 2,340 working-set papers were screened against IC4a–IC4d and EC1–EC4 using the LLM-assisted protocol described in Section 2.4. The screening was executed across five sequential batches of up to 500 papers each via the Anthropic Batches API, producing structured JSON outputs with decision, rationale, and matched criteria for each record.

Table 11: Title/abstract screening results.

Decision	Count	%
Include	111	4.7
Maybe	775	33.1
Exclude	1,454	62.2
Total	2,340	100.0

The `include` set (111 papers) and the `maybe` set (775 papers) together form the full-text review queue of 886 papers. The `exclude` decisions were driven primarily by EC1 (domain outside software development, principally generic business process management) and EC3 (theoretical algorithm proposals without SE evaluation).

3.4. Full-Text Screening

The 886 papers forwarded from the T/A phase were subjected to LLM-assisted full-text screening using the strict prompt described in Section 2.4 (decisions limited to `include`, `exclude`, or `pending`; `maybe` is not permitted at this stage). To maximise the number of papers assessable by the LLM, a multi-source abstract-enrichment cascade was executed prior to screening—combining DOI-based lookups via Semantic Scholar, OpenAlex, CORE, and Crossref, followed by title-based fallback lookups—raising abstract coverage from 16.6% to 72.6% (643 of 886 papers).⁶

⁶Papers were processed in order of a five-band priority scheme: Band A (91 papers, T/A `include` ≥ 2 ICs), Band B (20, T/A `include` 1 IC), Band C (39, T/A `maybe` + abstract), Band D (152, T/A `maybe` + IC signal, no abstract), Band E (584, T/A `maybe`, no IC signal, no abstract).

Table 12 summarises the outcome.

Table 12: Full-text screening outcomes (886 papers forwarded from T/A phase).

Outcome	Count	%
Included	169	19.1
Excluded	717	80.9
EC3 (algorithmic, no SE eval.)	(232)	(26.2)
EC1 (non-SE domain)	(205)	(23.1)
EC5 (full text inaccessible, post-audit active stratum)	(145)	(16.4)
EC2 (SW as implementation)	(110)	(12.4)
EC4 (out-of-scope secondary)	(60)	(6.8)
Total	886	100.0

Note: EC counts are non-exclusive — a single paper may trigger multiple ECs when more than one applies (e.g., a non-SE-domain paper whose contribution is purely theoretical). The EC5 count of 145 is the active post-audit stratum after re-screening the original historical EC5 band of 148 papers and reclassifying 3 of them to EC3. The sum of per-EC counts is $232 + 205 + 145 + 110 + 60 = 752$, which exceeds the 717 excluded papers by 35 (corresponding to papers triggering more than one EC). The percentages above are with respect to the 886-paper FT queue.

Among the 717 excluded papers, EC3 (algorithmic or theoretical contribution without SE evaluation, $n = 232$) and EC1 (domain outside software development, $n = 205$) jointly account for the largest share of FT-stage exclusions. EC5 (full text inaccessible despite all retrieval attempts, active post-audit $n = 145$) corresponds to the residual portion of the original 148-paper Band D/E stratum for which no open-access copy, institutional access, or author-provided preprint could be located; these are recorded in a dedicated traceability file and documented in the PRISMA flow as a named exclusion stratum. EC2 (software as implementation platform rather than analyzed process, $n = 110$) and EC4 (out-of-scope secondary studies, $n = 60$) account for the remainder.

Figure 1 presents the complete PRISMA 2020 flow diagram [11] of the selection process.

4. Results and Synthesis

4.1. Overview of Included Studies

The SLR is reported in two complementary tiers: (i) the **working-set tier**, comprising the 2,340-paper operational corpus that was deduplicated and processed through the LLM-assisted T/A and FT screening, yielding **169 confirmed primary studies**; and (ii) the **combined tier**, which adds the 3,807-paper deferred auxiliary corpus after a full T/A and FT pass on the same protocol, yielding 212 additional confirmed studies in the first auxiliary pass, plus 23 further confirmations from a second auxiliary pass over abstracts recovered through post-hoc enrichment (Section 6, External Validity), for a combined total of **404 raw confirmed primary studies** ($169_{\text{ws}} + 212_{\text{aux1}} + 23_{\text{aux2}}$), reduced to **340**

Abstract enrichment and LLM re-screening were applied to Bands A, B, C, and the subset of D/E that gained an abstract after enrichment.

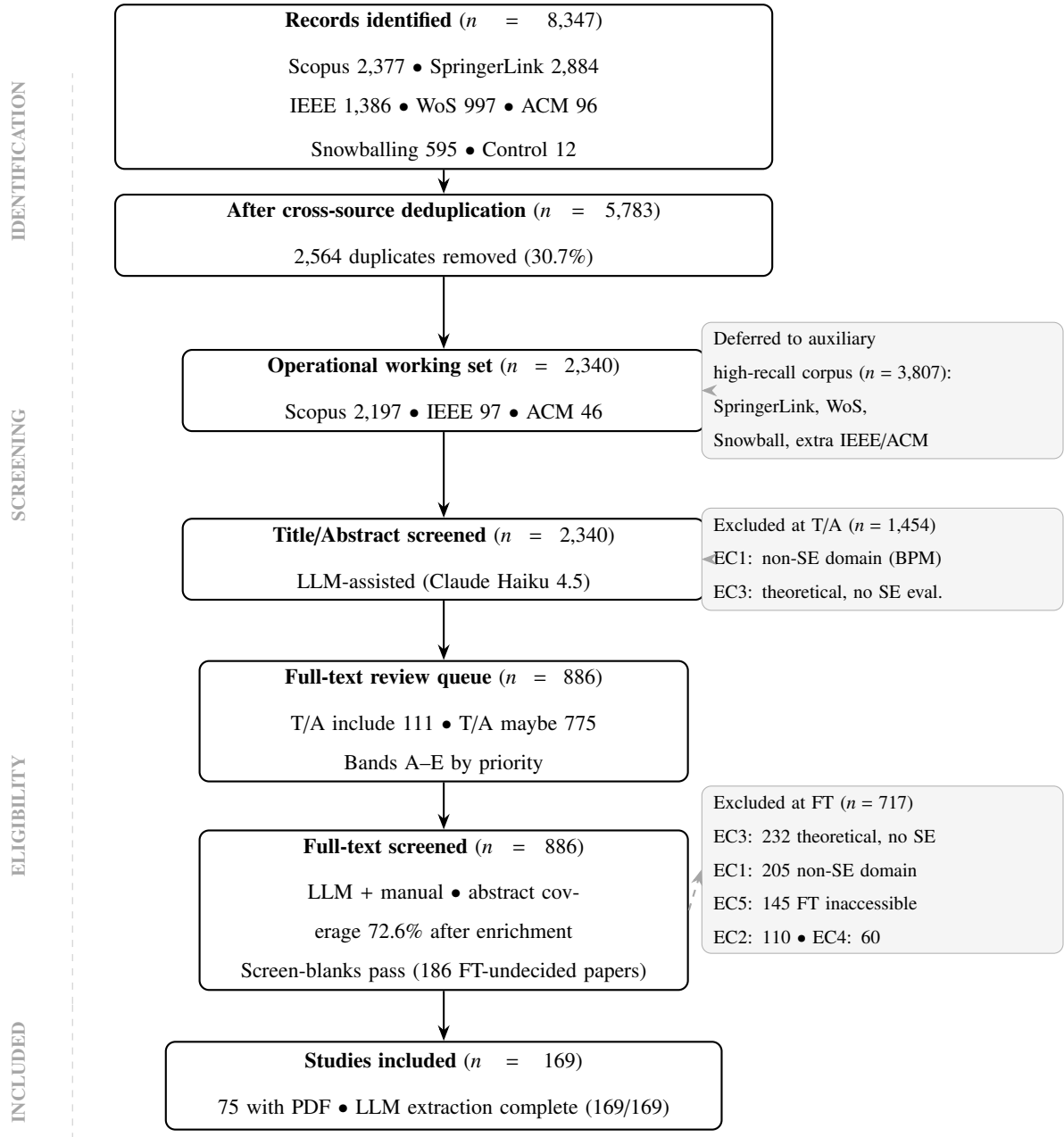


Figure 1: PRISMA 2020 flow diagram of the study selection process. Phase labels follow the four-stage structure: Identification, Screening, Eligibility, and Included.

distinct studies after cross-tier deduplication (63 duplicate records removed from the 381-study analytical freeze; 64 from the 404-tier freeze). The working-set tier is the narrative spine of this article; the combined-tier numbers are reported in parallel for findings F1–F5. Unless stated otherwise, combined-tier distributions refer to the **318-study deduplicated analytical subset** (259 pass reporting-quality $\geq 4/8$). The 169-study working-set tier spans 1995–2026 (publications dated 2026 that were already indexed on 12 April 2026 are retained), with a clear acceleration from 2014 onward (Figure 2). Sixty-nine percent of working-set papers (117/169) appeared in 2015–2026. The complete list of the 340 distinct confirmed studies is given in [Appendix A](#). A completed SEGRESS [1] reporting checklist is in [Appendix B](#).

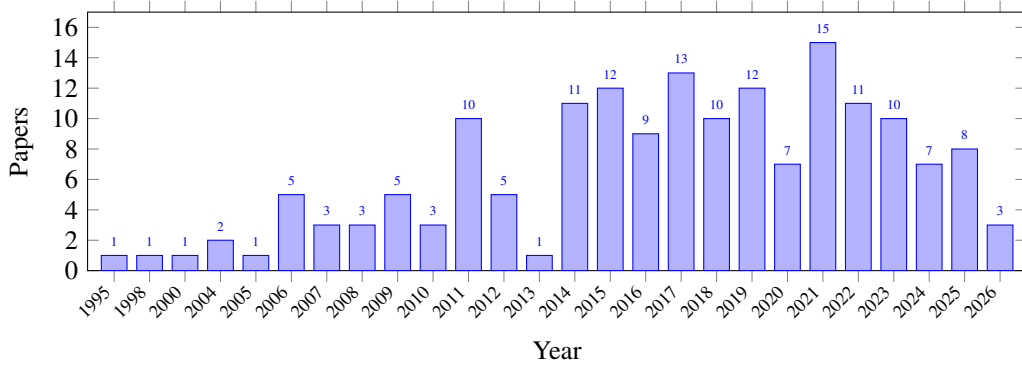


Figure 2: Distribution of confirmed studies by publication year ($n = 169$).

By document type, the majority are conference papers and journal articles. By source database, Scopus is the dominant contributor, followed by IEEE Xplore and ACM DL. Data extraction was completed for all 169 studies using a structured LLM-assisted pipeline (*claude-haiku-4-5-20251001*) operating on locally available PDFs (75/169, of which 73 were operationally coded in the triangulation subset) or abstract/title metadata (94/169).

4.2. Inclusion Criterion Coverage

Figure 3 shows the distribution of inclusion criteria among the 169 confirmed papers (criteria are non-exclusive; a paper may satisfy multiple).

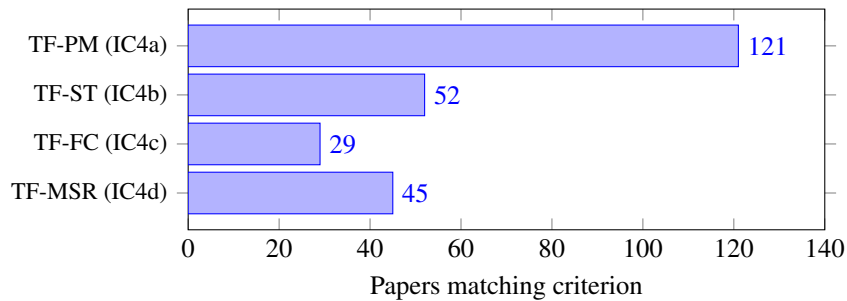


Figure 3: Inclusion criteria matched by the confirmed set ($n = 169$; non-exclusive; TF-PM/ST/FC/MSR = content criteria IC4a–IC4d).

TF-PM (process mining applied to SE artifacts) is the dominant criterion, matched by 121 papers (71.6%). TF-MSR (repository mining for process modeling) is second at 45 papers (26.6%), reflecting the MSR community’s overlapping contributions. TF-ST (stochastic modeling) appears in 52 papers (30.8%) and TF-FC (forecasting of process metrics) in 29 papers (17.2%). Notably, 68 papers (40.2%) match TF-PM only, without stochastic or forecasting components, providing direct empirical evidence for the “Descriptive Ceiling” identified in Finding F1 below.

4.3. Prevalent Exclusion Patterns

Among the 717 FT-excluded papers (Table 12), the largest exclusion reason is EC3 (algorithmic or theoretical contribution without software-engineering evaluation, $n = 232$), indicating a substantial body of process mining and stochastic modeling research that has not been empirically applied to software engineering contexts. EC1 (domain exclusively outside software development, $n = 205$) is the second-largest reason and consists primarily of generic business process management (BPM) studies that use the same terminology as the search strings but apply techniques to organizational processes unrelated to SDLC. EC5 (full text inaccessible, active post-audit $n = 145$; original historical stratum $n = 148$) closes the group of large strata, followed by EC2 (software as implementation platform rather than analyzed process, $n = 110$) and EC4 (out-of-scope secondary study, $n = 60$). EC counts are non-exclusive: the sum of per-EC counts ($232 + 205 + 145 + 110 + 60 = 752$) exceeds the 717 excluded papers by 35, corresponding to papers that triggered more than one EC simultaneously. The dominance of EC3 and EC1 directly supports Findings F3 and F4 below: process mining and stochastic methods are widely studied in the abstract or in non-SE domains, but their application to SDLC processes remains comparatively narrow.

4.4. Full-Text Extraction and PDF-Based Triangulation

Structured data extraction was performed on the full 169-study set using the extraction template defined in Table 8. For each study, the LLM extracted the research question, study type, research contribution, PM technique, stochastic technique, software artifact, software process, data source, dataset availability, tool used, main finding, limitations, and replication-package status. Title-and-abstract metadata was used as the primary input for studies without locally available PDFs (94/169); for the remaining 75/169 studies a locally accessible PDF was available and used to triangulate against the LLM-extracted fields. Of these 75 PDFs, 73 were fully coded in the operational triangulation spreadsheet used for the figures and positioning tables below.

The 73-paper PDF triangulation subset is therefore not a separate selection stage and does not override the PRISMA counts. It is used in this chapter as a verification layer for the LLM-extracted fields used in Findings F1–F5.

Figure 4 summarizes the operational triage profile of the 73 coded PDFs (54 confirm-include, 13 confirm-include with caveats, 6 confirm-exclude after full-text inspection); the formal selection decisions reported in the PRISMA flow (Figure 1) are unchanged.

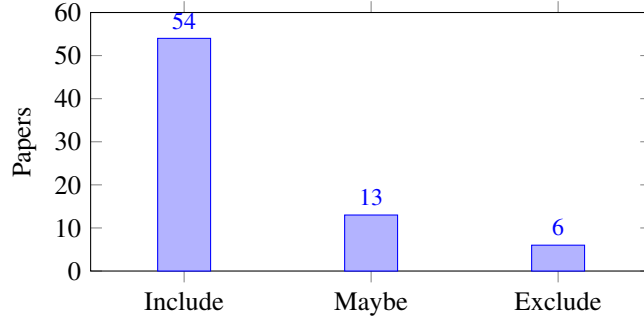


Figure 4: Operational decision distribution in the 73-paper PDF triangulation subset.

4.5. RQ1: Application Landscape

4.5.1. RQ1.1: SDLC Phases Covered

The structured extraction over the full 169-study set confirms a strongly uneven coverage of the SDLC. **Development activities** (coding, commits, pull requests, repository evolution) dominate at 133 papers (78.7%); **quality-related activities** (testing, defect handling, code review, process evaluation) appear in 74 papers (43.8%); **maintenance** appears in 38 papers (22.5%); **requirements engineering** in only 22 papers (13.0%); and **delivery/DevOps contexts** (CI/CD, build and release pipelines) in just 18 papers (10.7%). **Design and architecture** are not the primary focus of any extracted study, and only 3 papers (1.8%) center the analysis on project management.

Two conclusions follow. First, the literature is concentrated on phases where digital traces are abundant and already timestamped, such as commits, issues, test events, and pipeline executions. Second, the early SDLC phases remain structurally underrepresented, which weakens the ability of current approaches to support end-to-end forecasting from requirements to delivery. For PATHCAST, this supports a pragmatic focus on operational phases with reliable event data while leaving room for future extensions toward requirements and design logs.

4.5.2. RQ1.2: Event Log Sources and Construction

Across the 169-study set, event logs are constructed primarily from four source families: **version-control repositories** (Git/GitHub/SVN commits) appear in 84 papers (49.7%); **issue tracking systems** (especially Jira and GitHub Issues) in 37 papers (21.9%); **IDE interaction logs** in 22 papers (13.0%); and **CI/CD or build pipelines** in 15 papers (8.9%). The common pattern is not a native event log but rather a *derived* event log assembled from repository metadata, commit histories, issue transitions, and (where available) pipeline records.

The dominant construction problem is case correlation across tools. Studies frequently analyze one source in isolation or perform ad hoc joins between issues and commits, which limits reproducibility. This confirms that event-log construction is not a preparatory detail but a central methodological layer in software process mining. The finding directly justifies Stage 1 of PATHCAST as an explicit transformation pipeline rather than an implicit data-cleaning step.

A non-trivial fraction of studies report data sources only in coarse terms (*repository data*, *VCS history*, *open-source projects*) without naming specific projects, time windows, or extraction queries. Across the 169-study set, only 25 papers (14.8%) explicitly state that their dataset is publicly available and only 4 papers (2.4%) declare a replication package. This itself is evidence of incomplete reporting in the literature and provides direct empirical grounding for Finding F2 below.

A complementary body of work constructs Markov models directly from GitHub repository traces, bypassing explicit process discovery in favour of manual state-space definition. Jo et al. [12] modelled developer activities in GitHub repositories as Discrete-Time Markov Chains (DTMCs), finding that commits account for 62.73% of long-term activity, and employed probabilistic temporal logic to extract collaboration patterns relevant to project management. This approach was extended in [13], where Probabilistic Model Checking with Computation Tree Logic was applied across five repositories to characterise temporal dynamics, state-transition probabilities, and key behavioural drivers, yielding transparent and explainable project forecasts. Ortu et al. [14] complemented these structural analyses by combining Markov chains with social network analysis to identify fault-insertion and fault-fixing behavioural patterns across Apache Software Foundation projects, demonstrating that developer expertise significantly moderates defect introduction likelihood. Collectively, these studies confirm that SDLC event logs extracted from public repositories are sufficiently rich to support state-transition modelling, satisfying a foundational prerequisite for PATHCAST Stage 1. However, the event-to-state mapping adopted across these works remains unstandardised and ad hoc, representing a methodological gap that PATHCAST’s process mining discovery stage aims to address systematically.

4.6. RQ2: Techniques and Prediction

4.6.1. RQ2.1: Discovery Algorithms Applied to SDLC

The canonical origin of process mining applied to software development contexts can be traced directly to the foundational contributions of Cook and Wolf in the mid-1990s. In their seminal work, [15] introduced three automated process discovery methods—algorithmic grammar inference, Markov models, and neural networks—demonstrating that finite-state machine representations of software processes could be extracted automatically from event traces, substantially lowering the barrier to formal process modeling. This line of inquiry was consolidated in [16], which refined the event-based discovery methodology and established the conceptual vocabulary of inferring workflow structures from historical execution data, a paradigm that remains central to the field today. Notably, both contributions embedded stochastic reasoning—particularly Markov chain representations—within the discovery pipeline, anticipating the probabilistic extensions that would later characterise mature process mining frameworks. The translation of these principles into an explicit, named framework for software engineering contexts was accomplished by [17], who applied the ProM toolkit to source code management logs and demonstrated multi-perspective analysis spanning control-flow, informational, and organisational dimensions. Together, these three works constitute the seminal arc of TF-PM, establishing automated discovery from SDLC event logs as both technically feasible and analytically productive, and providing PATHCAST with its foundational methodological lineage.

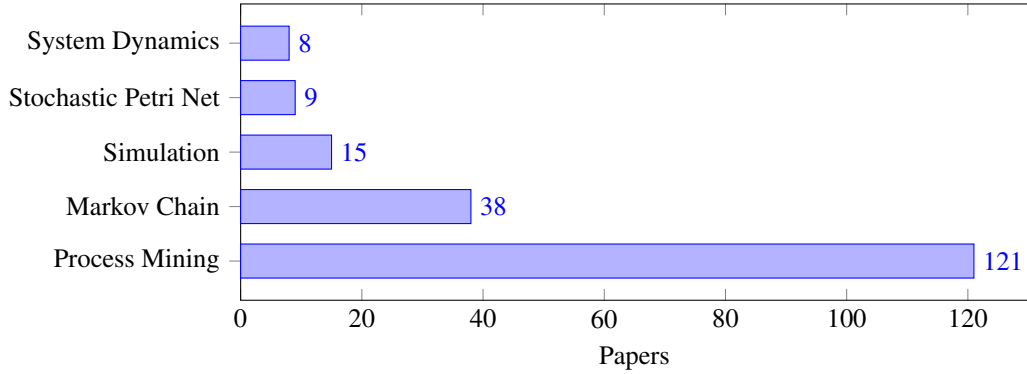


Figure 5: Main technique families identified in the 169-study confirmed set. Counts are non-exclusive: a paper may use multiple technique families.

The complete extracted set confirms that **process mining remains the dominant analytical family**: 121 of 169 studies (71.6%) match TF-PM in the form of discovery, conformance checking, or process-oriented repository analysis. At the technique level, the most frequently named PM algorithms across the 169 studies are the **inductive miner** (51 papers), **alpha miner** (45 papers), **conformance checking** (30 papers), and **heuristic miner** (22 papers); 11 papers use social network analysis on developer interactions. The literature is markedly richer in *process representation* than in probabilistic forecasting: only 29 papers (17.2%) match TF-FC (forecasting of process metrics), and the dominant contribution type is discovery or conformance checking rather than predictive analytics. This is the transition that PATHCAST targets: moving from discovered process structures to state-based probabilistic forecasting.

The stochastic-technique distribution at the same 169-study level shows the same fragmentation. Only 52 of 169 studies (30.8%) name an explicit stochastic technique: 38 papers (22.5%) use Markov chains, 15 (8.9%) use Monte Carlo simulation, 9 (5.3%) use stochastic Petri nets, and 8 (4.7%) use system dynamics. The remaining studies either do not adopt a stochastic component (110 papers) or place stochastic methods in an auxiliary role (*other*, 29 papers). The forecasting literature thus remains fragmented across adjacent technique families rather than consolidated around a single process-aware predictive paradigm, as shown in Figure 5.

4.6.2. RQ2.2: Predictive and Stochastic Techniques

Three distinct sub-families of stochastic techniques emerge from the confirmed literature. The first, **Markov-chain-based prediction**, encompasses models that represent software processes as discrete state transitions, including reinforcement learning agents grounded in Markov Decision Processes for pull-request outcome prediction [18], probabilistic risk quantification from repository data [19], defect removal effectiveness under code churn [20], agile state-space planning and tracking [21], release-time forecasting via generalised reliability models [22], and Stochastic Automata Networks for Waterfall team estimation [23]. The second sub-family, **stochastic Petri nets**, models concurrency and asynchrony in DevOps and CI/CD pipelines through non-Markovian specifications [24] and sensitivity analyses that identify bottleneck stages governing delivery probability [25, 26]. The third sub-family, **Monte Carlo simulation**, drives project-level forecasting in hybrid System Dynamics models [27], agile risk assessment

frameworks [28], cycle-time economic analysis [29], and the PRIMAD digital-twin architecture for SDLC [30].

Despite their methodological diversity, all three sub-families share a common parametrisation strategy that constitutes a structural limitation. Transition probabilities, firing rates, and distributional parameters are invariably derived from aggregate project statistics—historical defect counts, phase-level duration averages, or summary velocity metrics—rather than from transition frequencies extracted through systematic process mining of fine-grained event logs [31, 25, 27]. Consequently, model states remain anchored to coarse lifecycle phases and cannot reflect the richer, empirically grounded activity sequences that event logs encode. This observation constitutes **Finding F3**: *stochastic forecasting models for software development are parametrised from aggregate statistics rather than from mined process structures, confining their state spaces to coarse lifecycle granularity and decoupling model topology from observed execution behaviour*.

Three studies exemplify the productive convergence of process mining and stochastic modelling at the component level. Incerto [32] demonstrated that variable-length Markov chains outperform eighteen existing stochastic conformance-checking techniques across ten of eleven benchmark datasets, establishing that higher-order sequential dependencies in event logs can be captured without sacrificing computational tractability. López-Pintado [33] extended process-simulation fidelity by replacing deterministic resource-availability calendars with probabilistic counterparts discovered directly from event logs, showing that stochastic resource representations more accurately replicate empirical cycle-time distributions. Jalote [34] took a complementary perspective, modelling programmer task sequences as first-order Markov chains derived from IDE interaction logs and demonstrating that transferring the process patterns of high-productivity developers to average-productivity peers yields measurable productivity gains. Together, these contributions confirm that stochastic extensions of process models improve both explanatory and predictive validity; however, each study confines its stochastic layer to a single analytical concern without integrating multiple stages into a unified forecasting pipeline.

The closest published work to a fully integrated multi-stage architecture is PRIMAD, proposed by Guinea-Cabrera [30], which orchestrates pm4py-based process discovery, Akka actor-system simulation, and LightGBM supervised learning within an evolutionary Digital Twin framework for the SDLC. Nevertheless, PRIMAD remains at what may be characterised as Level 2 integration: validation is reported per component rather than across the assembled pipeline, no formal inter-stage contracts govern the handoffs between discovery, simulation, and learning modules, and the architecture incorporates no dedicated Monte Carlo forecasting layer capable of propagating uncertainty estimates end-to-end. The present review therefore concludes that the Level 3 integration niche—a formally contracted, end-to-end pipeline combining process discovery, absorbing Markov chain construction, Monte Carlo simulation, and residual correction—remains unoccupied in the extant literature.

Studies that combine process mining with forecasting objectives ($\text{TF-PM} \cap \text{TF-FC}$) tend to approach prediction through *process enhancement* rather than through explicit stochastic modelling. Gupta [35] demonstrated that combining inductive miner discovery with Markov chain-based predictive analytics enables identification of recurring failure patterns in software maintenance workflows. Pourbafrani et al. [36] presented SIMPT, a tool that extracts

process trees with time parameters from event logs and enables interactive what-if simulation; its successor work [37] further integrates generative AI with business process simulation to produce conformance-preserving what-if scenarios. Caldeira et al. [38] mine IDE interaction logs and show that process complexity metrics derived from ProM correlate moderately ($r \approx 0.43$) with software cyclomatic complexity, achieving predictive accuracies exceeding 92%, demonstrating the feasibility of process-derived features for product-quality prediction.

Taken together, this cluster confirms that PM and prediction are being connected in the SE literature, but always at the level of sequential application—PM output feeds a downstream model—rather than through a unified pipeline with formal contracts. Notably, none of these systems can express, as a first-class output, the probability that a given process instance will reach completion within k activity cycles—the precise forecasting primitive that distinguishes absorbing Markov chain formulations from ad hoc simulation runs. This gap motivates PATHCAST’s design, wherein the mined process structure feeds directly into an absorbing Markov chain whose transient-state absorption probabilities constitute a formally defined, queryable forecasting interface.

4.7. RQ3: Integration and Gaps

4.7.1. RQ3.1: PM and Stochastic Integration Level

Mapping the confirmed papers onto the L0–L3 integration scale reveals a clear stratification. The largest cluster, comprising papers that touch TF-ST and TF-FC without TF-PM—including stochastic Petri net models of CI/CD pipelines [25, 26], Markov-based project estimation [23, 21], and Monte Carlo risk simulation [28, 29]—operates almost entirely at **L1**, applying stochastic modelling to software processes without any process discovery layer. Papers spanning TF-PM and TF-ST, such as the VLMC conformance checker of [32] and the probabilistic resource-calendar discoverer of [33], reach **L2** by coupling discovered process structure with stochastic reasoning, yet neither closes the loop toward distributional schedule forecasting. Papers combining TF-PM with TF-FC, such as [35] and [38], use process mining descriptively to assess workflows but lack the stochastic machinery required for Monte Carlo distributional forecasts. The PRIMAD framework [30] is the closest antecedent, assembling elements from all three technique families—event-log discovery, simulation, and ML prediction—but its evaluation remains component-wise, placing it at L2 rather than a fully validated end-to-end pipeline. Notably, no paper in the reviewed corpus instantiates an **L3** architecture in which process mining discovery feeds an absorbing Markov chain, whose transient distributions drive Monte Carlo forecasting, with residuals corrected by a learned model, all validated on SDLC event logs. PATHCAST is sketched as a candidate L3 architecture, not as a claim of priority.

Quantitative evidence over the full 169-study set confirms the pattern. Only 1 of 169 papers (0.6%) jointly satisfies TF-PM, TF-ST, and TF-FC—that is, combines process mining, stochastic modeling, and forecasting in the same study—and even that single paper does not specify a formal end-to-end pipeline contract. Pairwise convergence is also low: $\text{TF-PM} \cap \text{TF-ST}$ (PM and stochastic) appears in 10 papers (5.9%), $\text{TF-PM} \cap \text{TF-FC}$ (PM and forecasting) in 7 papers (4.1%), and $\text{TF-ST} \cap \text{TF-FC}$ (stochastic and forecasting) in 18 papers (10.7%). **No paper** in the reviewed corpus implements an **L3** architecture. The literature has many partial bridges but no end-to-end process-aware forecasting

architecture for SDLC data. The operational coding of the 73-PDF triangulation subset is consistent with the full-set numbers and is summarized in Figure 6.

4.7.2. RQ3.2: Gap Analysis

Three structural observations emerge from the 169-study extraction. First, the field is **descriptively strong but forecastively weak**: 121 papers (71.6%) discover or assess process behavior, but only 29 (17.2%) estimate future paths or probabilistic completion distributions. Second, **evidence quality is uneven**: of the 169 confirmed studies, 69 (40.8%) are case studies, 34 (20.1%) are tool papers, 33 (19.5%) are theoretical, 13 (7.7%) are simulation-only, 10 (5.9%) are controlled experiments, 5 (3.0%) are surveys, and 5 (3.0%) fall in residual categories (mixed, framework, exploratory; sum: $69 + 34 + 33 + 13 + 10 + 5 + 5 = 169$). The literature is therefore method-heavy and applied-evidence light. Third, **reproducibility is poor**: only 25 of 169 papers (14.8%) declare their dataset publicly, and only 4 (2.4%) declare a replication package. The QA scoring in Table 7 confirms this pattern: only 33% of studies score 1 on QA7 (data/code availability), which is the single weakest QA criterion across the included set.

Taken together, these patterns explain why PATHCAST requires both a formal pipeline and an evaluation strategy grounded in real SDLC traces. The gap is not the absence of isolated techniques, but the absence of robust, end-to-end, process-aware forecasting workflows validated on heterogeneous software engineering data. The evidence-type distribution observed in the 73-PDF triangulation subset (Figure 7) is consistent with the full-set characterization above and is reported here as an analytical view, not as a separate selection layer.

5. Discussion

5.1. Research Gap Identification

The analysis of 2,340 screened records and 169 confirmed primary studies, together with the deduplicated 318-study combined analytical subset, supports five key findings. **Evidence bases**: F1–F5 counts for the working-set tier use all 169 confirmed studies; combined-tier percentages use the 318 deduplicated studies unless noted; synthesis percentages that exclude sub-threshold reporting-quality scores use 136/169 (working set) or 259/318 (combined). Structural scarcity claims (F3, F4) are interpreted conditionally on the human-validation Recall bounds reported in Section 6.

F1: Descriptive Ceiling. In the **working-set tier** (169 papers), 121 (71.6%) match TF-PM (process mining in SE) but only 29 (17.2%) advance to TF-FC (forecasting of process metrics). In the **combined analytical subset** (318 papers, deduplicated), the proportion shifts because the auxiliary corpus pulled in more stochastic-only and forecasting-only studies that did not employ explicit process mining: 160 (50.3%) match TF-PM, 100 (31.4%) match TF-FC, and 78 (24.5%) match TF-PM alone. In both tiers the dominant contribution type is discovery or conformance checking, not predictive analytics or distributional forecasting; the descriptive ceiling identified in the working-set tier persists in the combined tier.

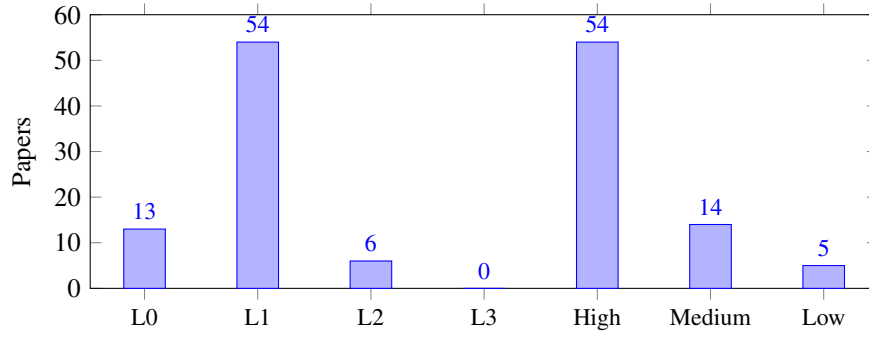


Figure 6: Operational integration levels and reading-priority distribution in the 73-paper PDF triangulation subset.

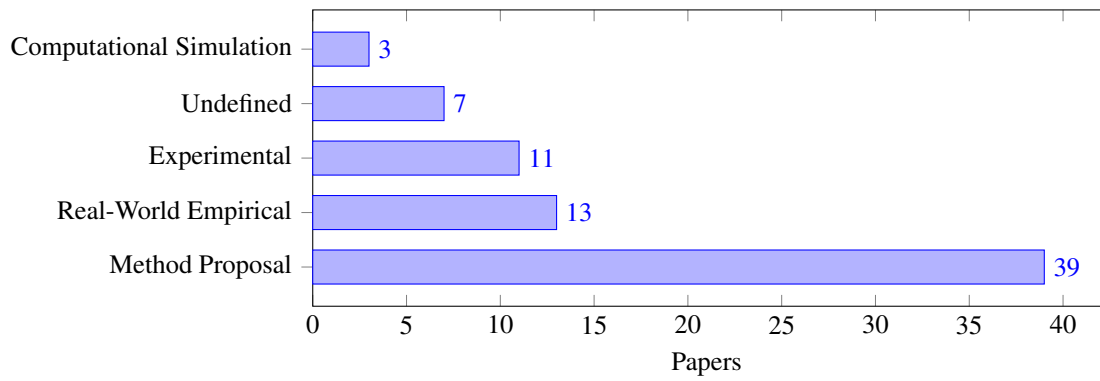


Figure 7: Evidence-type distribution in the 73-paper PDF triangulation subset.

F2: Event Log Construction Challenge. The reproducibility deficit is uniform across both tiers. Working-set tier: only 25 of 169 papers (14.8%) declare the dataset as publicly available and only 4 (2.4%) declare any replication package. Combined analytical subset: 44 of 318 papers (13.8%) declare a public dataset and 9 (2.8%) declare a replication package. The majority of studies that mention data collection procedures rely on ad hoc joins between version-control commits (112 of 318 papers, 35.2%), issue trackers (44, 13.8%), Jira (46, 14.5%), GitHub (43, 13.5%), and IDE interaction logs (37, 11.6%) rather than on a reusable construction protocol with explicit case-correlation semantics. The absence of standardized construction frameworks limits reproducibility and cross-study comparability and is the most common cause of QA3 (data source described reproducibly) failures observed in the QA scoring of the working-set tier.

F3: Stochastic Integration Gap. Working-set tier: 52 of 169 papers (30.8%) apply stochastic models (TF-ST); 38 (22.5%) Markov chains, 15 (8.9%) Monte Carlo, 9 (5.3%) stochastic Petri nets, 8 (4.7%) system dynamics; $TF-ST \cap TF-FC$ covers 18 papers (10.7%); $TF-PM \cap TF-ST$ covers 10 (5.9%). Combined analytical subset: the auxiliary corpus contributes a substantial number of stochastic-only papers not previously surfaced by the working-set queries; 152 of 318 (47.8%) match TF-ST (with 86 Monte Carlo, 67 Markov, 54 system dynamics, 15 SPN), 81 (25.5%) combine TF-ST with TF-FC (forecasting), and 13 (4.1%) combine TF-ST with TF-PM (process mining). Markov chains and Monte Carlo simulation are therefore used extensively in BPM and software reliability engineering, and the auxiliary expansion confirms a broader stochastic literature than the working-set tier alone suggested; nevertheless, the integration with process mining for SDLC forecasting remains minimal in both tiers (5.9% and 4.1% respectively for $TF-PM \cap TF-ST$). EC3 being the largest exclusion reason at FT in the working-set tier ($n = 232$, 26.2% of the 886-paper FT queue) reinforces the point: the stochastic modeling community has not converged on applying its techniques to software development process data.

F4: Pipeline Fragmentation. Studies apply techniques in isolation. The formal composition of a $PM \rightarrow Markov \rightarrow MC \rightarrow ML$ pipeline with inter-stage data contracts has not been proposed in the literature for SDLC contexts. The IC-combination analysis is robust across both tiers. Working-set tier (169 papers): the joint convergence $TF-PM \cap TF-ST \cap TF-FC$ appears in only 1 paper (0.6%), and that single paper does not specify a formal end-to-end pipeline contract; pairwise convergences are $TF-PM \cap TF-ST$: 10 (5.9%), $TF-PM \cap TF-FC$: 7 (4.1%), $TF-ST \cap TF-FC$: 18 (10.7%). Combined analytical subset (318 papers, deduplicated): $TF-PM \cap TF-ST \cap TF-FC$ remains 1 paper (0.31%), with pairwise convergences $TF-PM \cap TF-ST$: 13 (4.1%), $TF-PM \cap TF-FC$: 10 (3.1%), $TF-ST \cap TF-FC$: 81 (25.5%). These counts describe the LLM-confirmed, deduplicated corpus; given measured Lost Evidence at full-text screening (Recall = 0.246, 95% CI [0.179, 0.328], on the human validation sample, Section 6), scarcity findings should be read as conditional on the selection protocol rather than as a complete census. The closest antecedent in the corpus is PRIMAD [30], which assembles discovery, simulation, and learning components but validates each independently and leaves the unified pipeline unevaluated. Even when stochastic components are present, validation usually stops at process analysis or simulation without a down-

stream learning component.

PRIMAD is the only paper in the corpus that simultaneously satisfies $TF-PM \cap TF-ST \cap TF-FC$. Its scope is explicitly SDLC-oriented: it builds a digital twin from issue-tracking flows in JIRA to analyze and improve value streams in software organizations. Methodologically, it combines empirical process discovery (pm4py), a Python-based simulation core, an Akka actor system to model issues and resources, and supervised machine learning for context-aware prediction. The reported results are preliminary and component-wise rather than end-to-end: the paper shows promising independent results for the process-mining, simulation, and prediction components, but it does not yet validate a formally coupled $PM \rightarrow$ stochastic simulation $\rightarrow$ ML pipeline as a unified operational architecture, and remains the sole triple-family match among the 340 distinct studies reported here.

F5: Process-Derived Features Unexplored. ML-based prediction studies in the confirmed set rely overwhelmingly on traditional features (code metrics, commit characteristics, defect counts) rather than on features derived from process mining. Working-set tier: 32 papers whose research contribution centers on ML/prediction; only 1 explicitly couples ML with a named process-mining algorithm (alpha, inductive, heuristic, conformance, fuzzy, or Petri-net-based discovery). Combined analytical subset: the auxiliary corpus added a large number of ML/prediction papers (110 in the combined 318 set), but the rate of process-mining coupling remains low: only 4 of 110 (3.6%) jointly use a named PM algorithm. Conformance scores, rework counts, Markov-expected remaining time, and path entropy—features that are naturally produced by a process-mining stage—remain essentially absent from the SE prediction literature, in both tiers.

These five findings summarize the evidence gaps observed in the confirmed corpus. F1 corresponds to D1 (Descriptive Ceiling), F3 to D3 (Stochastic Integration Absence), F4 to D5 (Pipeline Fragmentation), and F5 to D4 (Feature Engineering Void). F2 provides empirical evidence for standardized event-log construction as a cross-cutting community need (RA1, Section 5.4).

5.2. PATHCAST as an Emerging Research Agenda

Table 13 summarizes integration levels in the 169-study working-set tier. PATHCAST is **not** evaluated in this mapping study; it is introduced below as a future-work hypothesis motivated by F1–F5.

PATHCAST names a candidate L3 architecture—process mining, absorbing Markov analysis, Monte Carlo forecasting, and ML enhancement composed for SDLC event data—as a **research agenda item** (RA2–RA4) rather than as an empirically validated contribution of this paper.

5.3. Proposed Taxonomy: SPMF

Based on the systematic analysis, this review proposes the Software Process Mining and Forecasting (SPMF) taxonomy with three dimensions:

Table 13: Integration levels in the 169-study working-set tier (family-based counts and 73-PDF operational subset).

Level	Description	Papers (169-set, family-based)	Papers (73-PDF subset, operational)	Representative examples
L0	Single analytical family (≤ 1 TF matched); no cross-technique chaining	95	13	Contextual or single-technique studies
L1	Two technique families matched; two families present, but no explicit reproducible pipeline contract	73	54	Stochastic models or ML without formal coupling
L2	Three technique families matched (TF-PM $\cap$ TF-ST $\cap$ TF-FC); partial bridge, incomplete pipeline	1	1	PRIMAD [30]
L3	Unified PM $\rightarrow$ Markov $\rightarrow$ MC $\rightarrow$ ML pipeline (proposed, not observed)	0	0	— (future-work hypothesis; Section 5.4)
Total	—	169	73	—

Analytical Depth classifies studies along four levels: L1 (Descriptive: discovery, conformance), L2 (Diagnostic: bottleneck, rework), L3 (Predictive: outcome, remaining time), and L4 (Forecasting: distributional, probabilistic).

SDLC Coverage maps which lifecycle phases are addressed: Requirements and Design, Development (coding, commits, PRs), Quality (testing, reviews, bugs), Delivery (CI/CD, release), and Operations (incidents, maintenance).

Technique Integration classifies methodological combination: Single (PM only, ML only, MC only), Dual (PM + ML, PM + Markov, Markov + MC), and Pipeline (PM $\rightarrow$ Markov $\rightarrow$ MC $\rightarrow$ ML).

5.4. Research Agenda

The gaps identified motivate four research directions:

RA1: Standardized Event Log Construction for SDLC. Motivated by F2: only 25/169 (14.8%) working-set papers declare public datasets; ad hoc joins dominate. Target: frameworks for transforming heterogeneous repository data into comparable event logs with cross-tool case correlation.

RA2: Process-Aware Stochastic Forecasting. Motivated by F3: TF-ST $\cap$ TF-FC reaches 18/169 (10.7%) in the working set but TF-PM $\cap$ TF-ST remains 10/169 (5.9%). Target: formal integration of mined transition structure with Markov/Monte Carlo forecasting.

RA3: Process-Derived Features for ML. Motivated by F5: only 1/32 ML-centric working-set papers couples a named PM algorithm with prediction. Target: systematic use of conformance, rework, and path-entropy features in SE predictors.

RA4: Empirical Benchmarking of Integrated Pipelines. Motivated by F4: a single distinct study (PRIMAD) satisfies $TF-PM \cap TF-ST \cap TF-FC$ among the 340 studies. Target: comparable OSS benchmarks with temporal hold-out for end-to-end PM–stochastic–ML pipelines; formal specification and evaluation of candidates such as PATHCAST remain future work by the authors.

5.5. *Implications for Practitioners*

Practitioners developing software measurement programs can extract three actionable insights from the 169-study confirmed set. First, the availability of structured event log data in issue tracking systems (Jira, GitHub Issues) is a prerequisite for process mining; organizations should invest in consistent workflow state management and transition logging before attempting process analysis. Second, open-source toolkits (PM4Py, ProM, Disco) suffice for discovery and conformance checking in most SDLC contexts, as suggested by the dominance of the inductive miner (51 papers) and alpha miner (45 papers) in the included studies’ tool usage patterns. Third, the stochastic extension (Markov chains for transition probability estimation, Monte Carlo for throughput forecasting) requires less infrastructure than typically assumed: the data already exists in the event logs; the challenge is methodological composition, not data collection.

6. Threats to Validity of the SLR

6.1. *Construct Validity*

The search strings may exclude studies that use non-canonical terminology. This threat is mitigated through three mechanisms: the complementary MSR string, the frontier stochastic-PM string, and snowballing. The 10/10 control paper recovery provides empirical evidence of high recall.

6.2. *Internal Validity*

LLM-based screening introduces the risk of systematic decision bias. This threat is addressed by five mechanisms, of which the first four were executed directly in the protocol and the fifth is encoded as a conservative closure rule rather than a mandatory human-arbitration step.

(i) **Structured outputs.** All screening responses use a strict JSON schema, enabling post-hoc analysis of decision patterns and rationale distributions.

(ii) **Author inspection of LLM include decisions.** Every paper retained at the FT stage was inspected by the first author against the PRISMA flow before being added to the 169-study confirmed set. This is a single-reviewer check without independent second-human reliability outside the blind sample below (SEGRESS item 23c).

(iii) **Independent human double-screening (gold standard).** A stratified random 20% sample of T/A ($n = 468$) and FT ($n = 177$) decisions was re-rated by the first author blind to the LLM’s decisions. Against the human rater as gold standard, strict binary Recall was 0.250 at T/A and 0.246 at FT (Lost Evidence 75% and 75%, respectively; Wilson 95% CIs in Table 15). All 54 T/A false negatives were LLM maybe rather than exclude decisions; treating maybe as a positive referral (LLM4SCREENLIT [5] R6) raises T/A Recall to 1.000 in this sample. Human–LLM κ was fair at T/A ($\kappa_{\text{binary}} = 0.335$) and slight at FT ($\kappa_{\text{binary}} = 0.122$); the low FT κ reflects a real criteria disagreement (the human rater included substantially more papers than the LLM), not merely prevalence effects (Table 14). On the 31 papers included by both raters, LLM QA matched the human rater on 68–100% per criterion; free-text extraction fields showed 0% exact-match agreement, so F1–F5 fields drawn from abstract-only auxiliary extraction are interpreted cautiously.

Table 14: Inter-rater agreement (Cohen’s κ) between the primary LLM screener (claude-haiku-4-5) and an independent human rater on a stratified random 20% sample. Interpretation thresholds follow Landis and Koch (1977) [9].

Stage	N	κ_{multi} (interpretation)	κ_{binary} (interpretation)
T/A	468	0.250 (fair)	0.335 (fair)
FT	177	0.122 (slight)	0.122 (slight)

Table 15: Human-vs-LLM confusion matrices on a stratified random 20% sample (human rater as gold standard; counts, not percentages). Recall and Lost Evidence (1–Recall) use Wilson 95% confidence intervals. The T/A row labelled “maybe→positive” treats LLM maybe decisions as positive referrals, matching LLM4SCREENLIT R6.

Stage	TP	FP	FN	TN	Recall	95% CI	Lost Ev.
T/A (strict)	18	4	54	392	0.250	(0.164, 0.361)	0.750
T/A (maybe→pos.)	72	105	0	291	1.000	(0.949, 1.000)	0.000
FT	31	3	95	48	0.246	(0.179, 0.328)	0.754

(iv) **Cross-model inter-rater agreement (Cohen’s κ).** A separate stratified random 20% sample of T/A ($n_{\text{theoretical}} = 468$ from the 2,340-paper working set) and FT ($n_{\text{theoretical}} = 177$ from the 886-paper FT queue) decisions was independently re-rated by a different model family (*claude-sonnet-4-6*) acting as a second rater. Of the 468 T/A papers in the sample, 420 produced a parseable decision from the verifier and entered the agreement computation; the remaining 48 papers returned a non-parseable response (typically because the abstract was absent, leading the verifier to refuse a categorical answer) and were excluded from the κ calculation as missing-at-random. All 177 FT sample papers produced a parseable decision. Both the multi-class κ (over the original include/exclude/maybe or include/exclude/pending categories) and the binary κ (collapsing the rater outputs to include versus not-include) are reported in Table 16. The binary κ is reported as a secondary signal: both binary κ values (0.695 for T/A and 0.694

for FT) exceed the substantial-agreement threshold $\kappa \geq 0.61$ [9], but two LLMs from the same provider may share systematic biases that inflate LLM–LLM agreement relative to an independent human check. The multi-class FT κ is lower (0.479, moderate) because the verifier returned pending on 25/177 FT papers where the primary screener committed to a binary decision.

Table 16: Inter-rater agreement (Cohen’s κ) between primary screener (claude-haiku-4-5) and verifier (claude-sonnet-4-6) on a stratified random 20% sample. Interpretation thresholds follow Landis and Koch (1977) [9].

Stage	N	P_o	P_e	κ_{multi} (interpretation)	κ_{binary} (interpretation)
TA	420	82.1%	50.2%	0.641 (substantial)	0.695 (substantial)
FT	177	76.3%	54.5%	0.479 (moderate)	0.694 (substantial)

The cross-model verification protocol was also extended to the auxiliary tier (FT stage, $n = 276$ stratified random 20% sample). On the auxiliary tier the verifier (Sonnet 4.6) returned pending on every paper that the primary screener (Haiku 4.5) had rated as include, yielding a multi-class $\kappa = 0.004$ and a binary $\kappa = 0.000$ ($P_o = 84.8\%$). For the auxiliary-tier T/A stratum ($n = 39$), $P_o = 97.4\%$ with $\kappa = 0.000$ — a base-rate/prevalence paradox, not evidence of disagreement between raters when one class dominates. Combined-tier findings therefore carry lower evidentiary weight than working-set-tier findings until auxiliary-tier human validation is completed.

Table 17: Auxiliary-tier inter-rater agreement (cross-model verification: Haiku 4.5 primary vs. Sonnet 4.6 verifier on a stratified 20% sample).

Stage	N	P_o^{multi}	P_o^{binary}	κ_{multi}	κ_{binary}
TA	39	97.4%	97.4%	0.000 (slight)	0.000 (slight)
FT	276	54.3%	84.8%	0.004 (slight)	0.000 (slight)

(iv) **T/A false-positive rate.** The LLM confirmatory pass at FT level (Round 1) rejected 33 of 108 T/A include papers (30.6%), primarily for EC3 (algorithm without SE evaluation). This false-positive rate is consistent with the T/A-screening false-positive rates reported by Khraisha et al. [8] for medical SLRs (in the 25–35% range, depending on domain specificity) and by Syriani et al. [7] for SE SLRs. False-positives are caught at the FT phase before any synthesis claim is made and therefore do not affect the F1–F5 evidence base. The Round 2 screening reprocessed 495 maybe papers with enriched abstracts and brought the confirmed set to its final value of 169.

(v) **Original closure of unresolved uncertainty (revised in resubmission).** In the original execution run, papers where metadata remained insufficient or verifier disagreement could not be resolved were closed under EC5-extended, biasing toward false negatives. Human validation (item iii) quantifies the cost: 75% Lost Evidence at FT under strict matching. The revised protocol routes maybe/pending to human review instead (Section 2.4).

6.3. External Validity

The review is limited to English-language peer-reviewed publications, potentially excluding relevant work in other languages or grey literature. The broad time span (1994–2025) and five-database coverage mitigate geographic and temporal bias. The deferred high-recall auxiliary corpus (3,807 records from SpringerLink, WoS, Snowball, and other low-precision queries; corresponds to the difference between the 5,783 unique deduplicated records and the 2,340-paper operational working set) provides a mechanism for expansion if gaps in the confirmed 169-study set are identified.

A four-stage sensitivity check was executed against the auxiliary corpus. First, a random 200-paper subsample was re-screened with the same T/A LLM protocol used for the working set; 12 of 200 (6.0%) returned a T/A include decision and 59 (29.5%) returned maybe, yielding a point-estimated 228 additional candidate includes if the auxiliary corpus were screened in full. Given that this estimate exceeded the working-set include count, the second stage was triggered: a **full-corpus T/A and FT screening pass** over all 3,807 auxiliary papers. The result is reported in Table 19: of the 3,807 auxiliary papers, 232 were T/A-included and 1,151 T/A-marked-maybe, of which 212 survived FT screening as include. Combined with the 169 working-set confirmations, the mapping study covers 404 raw confirmations (340 distinct after deduplication) in the combined tier; the 318-study deduplicated analytical subset is used for combined-tier F1–F5 tables.

The protocol then proceeded to a third stage: a multi-source abstract enrichment cascade (Semantic Scholar, OpenAlex, CORE, Crossref) was applied to the 880 auxiliary FT-pending papers, recovering an abstract for 285 papers (32.4%). The fourth stage re-ran the FT screening on these 285 enriched papers, confirming 23 additional includes, 246 excludes, and 16 still-pending. Under the conservative closure rule, those 16 residual abstract-insufficient papers are closed under EC5-extended rather than carried forward as a manual-validation queue. The combined confirmed set therefore becomes **404 raw primary studies** ($169_{ws} + 212_{aux1} + 23_{aux2}$), or **340 distinct studies** after cross-tier deduplication. The remaining 595 papers without abstract after enrichment are likewise closed under EC5-extended, rather than queued for manual retrieval, because they remain unclassifiable after the full multi-source enrichment cascade.

A second external-validity exposure concerns the original 148-paper EC5 stratum (full text inaccessible at first pass). To bound the residual recall risk, an automated open-access (OA) recovery pass was executed against four public discovery services (Unpaywall, Semantic Scholar Graph API, OpenAlex, CORE). The result is summarised in Table 21: 27 of the 148 EC5 papers (18.2%) returned at least one OA URL during the recovery pass; of those, 17 PDFs were successfully downloaded (11.5% of the stratum). A targeted audit of those 17 files found that 14 OA links resolved to unrelated papers or generic proceedings and therefore remained under EC5, while 3 papers were substantively re-screened and reclassified to EC3. The active PRISMA EC5 count after this audit is thus 145 rather than 148. The remaining 121 papers (81.8%) remain inaccessible and continue to be reported under EC5 in the PRISMA flow. A parallel OA-download pass over the 212 auxiliary includes recovered 49 PDFs (20.5% of 212; the lower yield reflects the publisher-paywall mix of the auxiliary corpus).

Table 18: Auxiliary-corpus sensitivity check. A random sample of n papers from the deferred auxiliary corpus was re-screened with the same T/A LLM protocol used for the working set.

Quantity	Value
Sample size	200 (seed=20260428)
Auxiliary corpus total	3807
Working-set confirmed includes	169
Sample T/A include rate	6.0% ($n_{\text{inc}} = 12$)
Sample T/A maybe rate	29.5% ($n_{\text{maybe}} = 59$)
Sample T/A exclude rate	64.5% ($n_{\text{exc}} = 129$)
Expected new includes (point estimate)	228
Expected new includes (upper bound, maybe→include)	1351
Sensitivity ratio vs. confirmed set (point)	135.2%
Sensitivity ratio vs. confirmed set (upper)	799.7%

Table 19: Auxiliary-corpus FT-screening outcome (papers that passed T/A on the 3,807-paper deferred corpus).

Outcome	Count	%
Forwarded from T/A	1383	100.0
Included (new)	212	15.3
Excluded	291	21.0
Pending / unparsed	880	63.6
Combined SLR set (working-set + auxiliary)	381	—

Table 20: Re-FT screening of aux-pending papers that gained an abstract through the post-hoc enrichment cascade.

Outcome	Count	%
Re-FT include	23	8.1
Re-FT exclude	246	86.3
Re-FT pending	16	5.6
Total re-FT	285	100.0

Table 21: EC5 recovery attempt against four open-access discovery APIs.

Source	Recovered	% of EC5
Unpaywall	6	4.1%
Semantic_Scholar	6	4.1%
Openalex	6	4.1%
Core	20	13.5%
Any source	27	18.2%

6.4. Conclusion Validity

Synthesis findings F1–F5 are derived from the structured data extraction applied to two extraction layers: (i) the full 169-study working-set extraction (Section 2.6; Table 8), partly triangulated against the 73-PDF subset; and (ii) the 212-study auxiliary-tier extraction performed via abstract only. Both extractions feed the IC-combination counts and the technique distributions reported under F1–F5. The QA-based filtering reported in Table 7 retains 136 of the 169 working-set studies (80.5%); QA scoring was subsequently extended to the auxiliary tier, retaining 179 of 212 (84.4%, mean 5.09 ± 1.43). The combined 318-study deduplicated subset retains 259 papers (81.4%, mean 5.00 ± 1.42) above the $\geq 4/8$ threshold; full combined-QA results are reported in Table 22.

Table 22: Quality assessment combined across working-set and auxiliary tiers ($n = 381$).

Tier	<i>N</i>	Mean	SD	$\geq 4/8$ (pass)	$< 4/8$ (fail)
Working-set	169	4.88	1.40	136	33
Auxiliary	212	5.09	1.43	179	33
Combined	381	5.00	1.42	315	66

A residual conclusion-validity threat is the LLM-based extraction of PM-technique, stochastic-technique, and SDLIC-phase fields. To mitigate this, the 73-PDF triangulation subset was used to spot-check the extracted fields against the corresponding full texts; no systematic disagreement was observed in the spot-check, but human re-extraction on the stratified sample ($n = 31$ papers included by both human and LLM) shows 0% exact-match agreement on free-text fields and 29–81% on categorical fields (Section 6, item iii).

6.5. Reproducibility and Use of AI

In line with current Elsevier and Springer-Nature editorial policy, this review explicitly declares the use of AI tools in its conduct. The primary screener at the title/abstract and full-text stages is *claude-haiku-4-5-20251001* (Anthropic), invoked through the Anthropic Message Batches API for screening and through the synchronous Messages API for QA

scoring. The cross-model inter-rater verification described above uses *claude-sonnet-4-6* (Anthropic) as the second rater. AI tools are **not** listed as authors of this work and are responsible for none of the editorial judgements; final selection, QA threshold, taxonomy design, and synthesis claims are the responsibility of the authors.

A complete replication package is deposited at Zenodo ([DOI 10.5281/zenodo.21939600](https://doi.org/10.5281/zenodo.21939600)). It contains the search strings for all five databases; deduplication and enrichment scripts; screening and QA prompts with JSON schemas; per-stage decision tables; reporting-quality and extraction tables for both corpus tiers; LLM–LLM and human–LLM agreement reports; the 340-study included list; a completed SEGRESS checklist; and a PRISMA flow snapshot. File-level layout is documented in the package README. The deposit is intended to be self-contained: a reviewer can re-run the LLM-assisted protocol on the deposited working set under the same model identifiers.

7. Conclusion

This article presented a systematic mapping study of the intersection of process mining, stochastic modeling, and machine learning applied to software process forecasting, reported against SEGRESS [1]. No review protocol was pre-registered; this is disclosed as a limitation. The review executed three complementary search strings across five databases (executed 12 April 2026; eligibility window 1994–2025), yielding 8,347 records, 5,783 after deduplication, and a 2,340-paper operational working set for systematic screening. Title and abstract screening on the working set using a LLM-assisted protocol produced 111 candidate include decisions and 775 maybe decisions (886 papers for full-text review).

Full-text screening combined LLM-assisted passes (Rounds 1 and 2, with cascade abstract enrichment raising coverage from 16.6% to 72.6%, i.e., 643/886 papers) with a final *screen-blanks* pass that processed the 186 papers that still lacked an FT decision after the earlier rounds. In the final screening artifact, 238/886 papers still lack an abstract field, but all have a recorded FT decision. Papers whose full text remained inaccessible despite all retrieval attempts were closed under EC5; after the OA recovery audit, 145 papers remain in that stratum. The working-set result is **169 confirmed primary studies** and 717 FT excluded (EC3: 232, EC1: 205, EC5: 145, EC2: 110, EC4: 60), with zero pending decisions. Data extraction for all 169 studies was completed using a structured LLM-assisted pipeline operating on locally available PDFs (75/169, with 73 operationally coded in the triangulation subset) or abstract/title metadata (94/169).

A second tier of the protocol then applied the same T/A and FT screening to the previously deferred 3,807-paper auxiliary corpus, yielding 232 T/A includes and 1,151 T/A maybes; FT screening of the 1,383 forwarded papers produced 212 additional confirmed primary studies (first auxiliary pass), 291 excludes, and 880 pending. A subsequent re-screening pass over 285 abstract-enriched auxiliary papers confirmed 23 further studies (second auxiliary pass), bringing the raw combined total to **404 primary studies**, or **340 distinct studies** after cross-tier deduplication. The deduplicated 318-study analytical subset supports combined-tier F1–F5 tables; structural scarcity claims are interpreted conditional on human-validation Recall bounds (Section 6).

Analysis of the working-set tier (169 studies) and the 318-study deduplicated combined subset supports five key findings. First, the field exhibits a descriptive ceiling: 71.6% of confirmed working-set papers (50.3% combined) apply process mining to software artifacts (TF-PM), but only 17.2% advance (working-set; 31.4% combined) to forecasting of process metrics (TF-FC) (F1). Second, event log construction from heterogeneous repository sources is a pervasive challenge (F2). Third, the integration of stochastic models with process mining for SDLC forecasting remains limited: 10.7% working-set / 25.5% combined for $TF-ST \cap TF-FC$ (F3). Fourth, only one distinct study (PRIMAD) satisfies $TF-PM \cap TF-ST \cap TF-FC$ among the 340 studies (F4). Fifth, ML studies in SE rarely incorporate process-derived features (F5).

These gaps motivate the SPMF taxonomy and research agenda RA1–RA4 (Section 5.4). PATHCAST names a candidate integrated pipeline as future work by the authors, not as an empirically validated contribution of this mapping study.

Methodologically, cross-model verification achieved substantial LLM–LLM κ (0.695 T/A; 0.694 FT), but independent human double-screening showed fair/slight human–LLM κ and 75% measured Lost Evidence at FT—the primary validity evidence for selection. Reporting-quality scoring covers all 169 working-set studies (136/169 above $\geq 4/8$). The replication package is openly available at Zenodo (DOI 10.5281/zenodo.21939600).

Appendix A. Included primary studies

The 340 distinct primary studies confirmed after cross-tier deduplication are not tabulated in this PDF. The machine-readable list, with year, authorship when available, title, DOI or internal identifier, corpus origin, technique family (TF-PM/ST/FC/MSR = IC4a–IC4d), and reporting-quality score, is file `included_studies_340.csv` in the replication package (DOI 10.5281/zenodo.21939600). Origin WS has 167 rows because two of the 169 working-set confirmations were duplicates of auxiliary records and were removed in cross-tier deduplication. Two auxiliary records (1991 and 1992) predate the IC1 publication window (1994–2025); they are retained as historical software process-modelling studies. Publications dated 2026 that were already indexed on 12 April 2026 are retained, as declared in the search protocol. Frequency claims in the main text can be traced to named studies via that file (SEGRESS item 17).

Appendix B. SEGRESS reporting checklist

Table B.23 maps each SEGRESS item to the location in this manuscript. Items marked N/A are those SEGRESS declares not required for mapping studies (no synthesis of primary-study outcomes).

Table B.23: SEGRESS reporting checklist for this systematic mapping study. Locations refer to the revised manuscript. Status: R = reported; P = partial; N/A = not required for mapping studies under SEGRESS.

Item	Requirement (mapping)	Where addressed	St.
1	Title identifies the report as a systematic mapping study and states the topic.	Title page: “A Systematic Mapping Study with an LLM-Assisted Protocol”.	R
2	Structured abstract: background, objective, methods, results, limitations, conclusion.	Abstract (incl. a Limitations clause and human-validation statement).	R
3	Rationale: why the review is needed; gap vs prior secondary studies.	Introduction: states that no prior mapping covers the joint $PM \times stochastic \times forecasting$ intersection.	R
4	Objectives / research questions.	Table 1; Section 2.1.	R
5	Eligibility criteria, date/language restrictions, handling of on-topic prior reviews.	Table 5 (IC1–IC3 publication filters; IC4a–IC4d content; EC1–EC5). Window Jan 1994–Dec 2025. On-topic secondary studies would be used to motivate the work (EC4 excludes off-topic reviews).	R
6	Information sources with search execution dates.	Table 3; all databases searched on 12 April 2026.	R
7	Full search strategy (strings, snowballing, completeness checks).	Section 2; per-database strings in the replication package; snowballing from the control set (Table 4). Control-set recovery 10/10 is a weak recall bound, not a completeness guarantee.	R
8	Selection process: phases, number of assessors, disagreement handling, automation, agreement.	Sections 2.4, 6: one LLM screener; one human rater on a 20% blind sample; original EC5-extended closure of uncertainty (revised protocol treats maybe/pending as human referrals). κ , Recall, Lost Evidence in Tables 14–15.	R

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Table B.23 — continued from previous page

Item	Requirement (mapping)	Where addressed	St.
9	Data-collection process: reviewers, independence, automation.	LLM extraction (Haiku) on abstracts/PDFs; human re-extraction on $n = 31$ dual-includes; no independent second human extractor on the full set (SEGRESS 23c limitation).	R
10a	Outcomes for which data were sought.	N/A: mapping study does not synthesise primary-study outcomes.	N/A
10b	Non-outcome variables and classification schemes.	Extraction schema: PM technique, stochastic technique, SDLC phase, dataset source, IC combination, reporting-quality scores (QA1–QA8).	R
11	Risk-of-bias methods (optional for mapping studies).	Reporting-quality checklist (Dybå & Dingsøyr), not RoB; purpose stated as a sensitivity filter on F1–F5, not a validity judgement. Scored by LLM; human check on $n = 31$. Threshold $\geq 4/8$.	R
12	Effect measures.	N/A (no outcome synthesis).	N/A
13 / 13a	Methods to analyse study characteristics; which studies enter each analysis.	Section 5.1: working-set $N = 169$; combined analytical $N = 318$ (deduplicated); QA-passed 259/318.	R
13b	Data preparation for outcome synthesis.	N/A (mapping).	N/A
13c	Methods for tables, graphs, maps of study characteristics.	Frequency tables, year plot, IC-combination counts, SPMF taxonomy, integration-level Table 13.	R
13d	Methods used to synthesise primary-study outcomes.	N/A (mapping).	N/A
13e	Sensitivity of the auxiliary corpus (not outcome synthesis).	Section 6, Table 18.	P
13f	Heterogeneity of primary-study outcomes.	N/A (mapping).	N/A

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Table B.23 — continued from previous page

Item	Requirement (mapping)	Where addressed	St.
14	Reporting-bias / publication-bias methods.	N/A for mapping studies under SEGRESS.	N/A
15	Certainty (GRADE) methods.	N/A for mapping studies.	N/A
16a	Selection results, ideally a PRISMA flow; agreement statistics.	Section 4 funnel narrative; PRISMA figure; LLM–LLM and human–LLM κ ; Recall / Lost Evidence. Raw 404 $\rightarrow$ 340 after deduplication.	R
16b	Near-misses (optional for mapping).	EC3 is the modal FT exclusion; a full near-miss citation list is not included (optional).	P
17	Cite each included study and present characteristics.	Full list and per-study characteristics: replication package <code>included_studies_340.csv</code> (DOI 10.5281/zenodo.21939600); manuscript pointer in Appendix A .	R
18	Risk-of-bias results per study (optional for mapping).	QA totals in <code>included_studies_340.csv</code> ; per-criterion scores in the replication package; working-set summary in Section 2.5.	R
19	Results of individual studies (outcomes).	N/A (mapping).	N/A
20a	Analyses of study characteristics addressing each RQ.	Sections 4–5.1 (RQ1–RQ3, F1–F5).	R
20b–20d	Statistical outcome syntheses, sensitivity, heterogeneity.	N/A (mapping).	N/A
21	Reporting-bias results.	N/A (mapping).	N/A
22	Certainty of evidence.	N/A (mapping).	N/A
23a	Interpretation in the context of other evidence.	Discussion / research agenda (Section 5.4).	R
23b	Limitations of the included evidence.	Optional for mapping; extraction-basis and Lost Evidence discussed in Section 6.	R
23c	Limitations of the review process.	Section 6: single human rater outside the blind sample; LLM–LLM non-independence; FT Recall 0.246; no pre-registration.	R

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Table B.23 — continued from previous page

Item	Requirement (mapping)	Where addressed	St.
23d	Implications for practice, policy, future research.	Sections 5.5, 5.4 (RA1–RA4). Mapping studies: future research is the required emphasis.	R
24a	Registration (or statement that the review was not registered).	Conclusion: no protocol was pre-registered (disclosed limitation).	R
24b	Protocol availability (optional for mapping).	No standalone protocol deposit; methods are specified in Section 2 and the replication package.	P
24c	Protocol amendments (optional for mapping).	Revised maybe/pending handling documented as a post-submission protocol change (Section 2.4).	R
25	Sources of financial / non-financial support.	Funding statement (CAPES Finance Code 001).	R
26	Competing interests.	Declaration of competing interest.	R
27	Public availability of data, code, and other materials.	Zenodo DOI 10.5281/zenodo.21939600; 340-study list included_studies_340.csv (pointer in Appendix A). Record is open access (LLM4SCREENLIT R7).	P

Source. Item definitions follow Kitchenham, Madeyski and Budgen, SEGROSS, *IEEE Trans. Softw. Eng.*, 49(3):1273–1298, 2023, doi:10.1109/TSE.2022.3174092.

CRediT authorship contribution statement

Rodrigo Almeida de Oliveira: Conceptualization, Methodology, Software, Investigation, Data curation, Formal analysis, Visualization, Writing – original draft, Writing – review & editing. **Juliano de Paulo Ribeiro:** Methodology, Validation, Investigation, Writing – review & editing. **Edson Emilio Scalabrin:** Conceptualization, Methodology, Supervision, Writing – review & editing, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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Declaration of generative AI in the writing process

During the conduct of this systematic literature review, the authors used *claude-haiku-4-5-20251001* (Anthropic) as the primary LLM screener at the title/abstract and full-text stages and as the LLM scorer for the quality-assessment rubric, and *claude-sonnet-4-6* (Anthropic) as the cross-model verifier for inter-rater agreement. All decisions were inspected by the authors; the LLM tools are not listed as authors and bear no responsibility for editorial judgements. Full prompts, JSON schemas, and per-paper decisions are deposited in the replication package ([Zenodo DOI 10.5281/zenodo.21939600](https://zenodo.org/doi/10.5281/zenodo.21939600)). The authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Data availability

A full replication package is openly available at Zenodo ([DOI 10.5281/zenodo.21939600](https://zenodo.org/doi/10.5281/zenodo.21939600)). The package contains: search strings for all five databases, deduplication and enrichment scripts, screening prompts and JSON schemas, the QA prompt, the cross-model κ verification scripts, all decision CSVs at every pipeline stage, the QA-scored 169-study and 212-study tables, the structured extraction tables (working-set and auxiliary), the inter-rater agreement reports (LLM–LLM and human–LLM), the 340-study included list (`included_studies_340.csv`), a completed SEGRESS checklist, and a PRISMA flow snapshot.

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